

Undergraduate Medicine

Pulmonology Study Pack

CharaNas Medicine Bot — Specialty Notes

Expanded notes with original schematic figures for revision. These are educational drawings inspired by standard undergraduate teaching themes, not copied textbook plates. Always follow your faculty curriculum and latest guidelines.

Section	Focus
1	Respiratory physiology basics
2	Pneumonia & severity
3	Asthma & COPD
4	PE, pneumothorax, TB
5	Checklist & books
6	Quick pearls from the bot notes
7	Recommended book sources

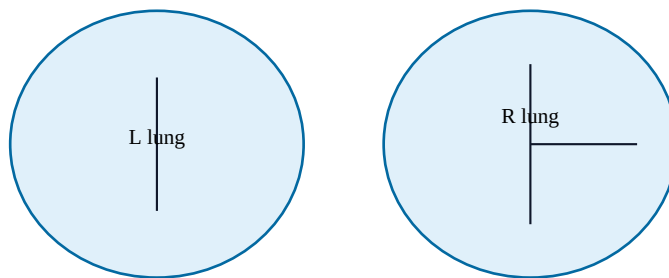
1. Respiratory physiology basics

Gas exchange requires ventilation, perfusion, and diffusion. Hypoxemia mechanisms: hypoventilation, V/Q mismatch, shunt, diffusion limitation, low inspired O₂.

Obstructive diseases reduce airflow (asthma, COPD); restrictive patterns reduce volumes (ILD, neuromuscular, kyphoscoliosis).

ABG interpretation: look at pH, PaCO₂, HCO₃, PaO₂ — decide acidosis/alkalosis and respiratory vs metabolic primary process.

Figure: Lung fields & lobar idea (schematic)



Lobar pneumonia consolidates a lobe; listen and correlate with CXR

Schematic figure for learning — verify details in your recommended textbooks.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about pulmonology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

2. Pneumonia & severity

CAP classic organism: *Streptococcus pneumoniae*. Assess severity with tools such as CURB-65 and clinical judgment.

Investigations: CXR, O2 sats, bloods, cultures when septic/hospitalized. Antibiotics according to local guidelines and risk for resistant organisms.

Complications: parapneumonic effusion, empyema, abscess, respiratory failure, sepsis.

Revision prompts

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- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

3. Asthma & COPD

Asthma: reversible obstruction and hyperresponsiveness; wheeze, dyspnea, cough; peak flow variability. Acute severe asthma is life-threatening — escalate early.

COPD: progressive largely irreversible obstruction, usually smoking-related; exacerbations with infection/irritants. Oxygen targets are often controlled (e.g., 88–92%) in CO₂ retainers — follow local protocol.

Inhaler technique teaching is as important as the prescription.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about pulmonology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

4. PE, pneumothorax, TB

PE: sudden dyspnea/pleuritic pain with risk factors — Wells score guides D-dimer vs CT PA pathway.

Tension pneumothorax: shock, tracheal deviation, absent breath sounds — immediate decompression, do not delay for CXR.

TB: chronic cough, night sweats, weight loss, hemoptysis — infection control + microbiologic testing (AFB/NAAT) as available.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about pulmonology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

5. Checklist & books

Checklist: RR, SpO₂, work of breathing, auscultation, CXR, ABG if tiring/hypercapnia risk, VTE risk, smoking history.

Books: West's Respiratory Physiology; Harrison's respiratory chapters; Crofton & Douglas.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about pulmonology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

6. Quick pearls

High-yield reminders packaged with this specialty in the CharaNas bot:

1. CAP: *S. pneumoniae* most common; assess CURB-65 severity.
2. Asthma: reversible obstruction; COPD: largely irreversible.
3. PE: sudden dyspnea/pleuritic pain post-risk — Wells + CT PA pathway.
4. Tension pneumothorax: shock + tracheal deviation — immediate decompression.
5. TB: chronic cough, night sweats, weight loss — AFB/GeneXpert as available.
 - Link symptoms → anatomy/physiology → investigation → first management step.
 - Write one OSCE-style explanation for a common presentation in this specialty.
 - After reading, do the bot Short MCQ and Case-based Question for active recall.

7. Recommended book sources

Standard undergraduate references commonly used internationally. Use the edition your college recommends. Figures in commercial textbooks are copyrighted; use library/licensed access for full plates and photographs.

#	Reference
1	West's Respiratory Physiology
2	Harrison's — Respiratory Medicine
3	Crofton and Douglas's Respiratory Diseases

How to study with this PDF: read one section → sketch the figure from memory → answer bot MCQs/cases → review mistakes → revisit the matching section.

Disclaimer: educational aid only; not a substitute for clinical guidelines, faculty teaching, or licensed textbooks.